Robert Lehr

Austin, Texas | github.com/robert-z-lehr

Systems & Transportation Engineering | Data, AI-Assisted Tool Development, Urban Systems

PROFILE

Systems-oriented engineering graduate student with experience in transportation engineering, AI-assisted technical tool development, urban sustainability, data analysis, and simulation. Focused on converting messy real-world engineering processes into reproducible workflows, usable tools, and defensible decisions.

EDUCATION

The University of Texas at Austin

Graduate Student, Civil Engineering / Sustainable Systems Engineering | 2024-Present

- Graduate research in urban systems, pedestrian heat exposure, transportation, and infrastructure decision-making.

The University of Texas at Austin

M.S., Sustainable Design | 2020 | GPA 3.56

Southwestern University

B.A., Mathematics | 2015 | GPA 3.38

EXPERIENCE

Lee Engineering, LLC - Transportation Engineering Intern

Dallas, TX | Jun 8-Aug 19, 2026

- Developed a deterministic internal automation tool for a time-intensive TIA data workflow, using Claude as a coding assistant while working with multiple engineers to define requirements, review outputs, and improve reliability.
- Designed the workflow so engineering calculations and outputs remain reproducible and auditable rather than relying on generative-model uncertainty.
- Pre-deployment estimate: potential to save 500+ staff hours annually and shorten project turnaround if deployed at expected usage; the estimate has not yet been validated through realized annual savings.
- Audited and iteratively improved developing internal AI-use guidelines, helping translate emerging AI capabilities into practical engineering safeguards and use cases.
- Shadowed engineers during four TIA site visits and assisted with field-value validation, connecting site observations to the data process being automated.

AI Training & Evaluation - Contract Projects

GlobalLogic / partner organizations including Digitive | 2024-2026

- Completed AI training and evaluation work across summer 2024 and summer 2025 engagements.
- Successfully completed a short-term Digitive contract in 2026 after being contacted for additional work.
- Continues to receive periodic contract invitations as a previously validated worker.

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Resume - Expanded Version

RESEARCH & APPLIED TECHNICAL WORK

The University of Texas at Austin - Graduate Research

Austin, TX | 2024-Present

- Investigates whether transportation heat-safety analysis changes when pedestrian exposure is represented with solar radiation, shade, and urban geometry rather than ambient air temperature alone.
- Integrates field environmental measurements, pedestrian observations, GIS, crash records, weather data, solar position, and local shade/geometry information.
- Develops structured field, validation, and data-management protocols for reproducible analysis.

Selected Applied Projects

Academic and independent portfolio

- Kestrel WBGT Demo: browser-based environmental-data demonstration supporting applied heat-exposure work.
- Community Voice at the Corner: place-based civic-feedback concept using geo-tagged input during routine urban dwell time.
- Builds browser-based analytical and decision-support prototypes that translate technical ideas into interactive workflows.

TECHNICAL SKILLS

Programming & data: Python, R, MATLAB, HTML/CSS/JavaScript, Git/GitHub, data visualization, statistical analysis.

Engineering & spatial: GIS/ArcGIS, transportation and environmental field-data workflows, simulation and optimization.

Applied methods: technical tool development, workflow automation, systems thinking, model validation, reproducible analysis.

PUBLICATION & RECOGNITION

- Fumiko Futamura and Robert Lehr, "A New Perspective on Finding the Viewpoint," Mathematics Magazine 90(4), 267-277, 2017.
- MAA Carl B. Allendoerfer Award, 2018, for coauthorship of an outstanding Mathematics Magazine paper.

SELECTED LEADERSHIP

Prior undergraduate leadership included President, Pi Mu Epsilon; Vice President, Mathematics Club; founder of the Think Series; and lead alto saxophone in the Southwestern University Jazz Band.